

NAVJOT SINGH CHAHAL

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Education

University of Pennsylvania, Philadelphia, USA **June 2026**
Master of Science, Electrical Engineering (Department of Electrical Systems Engineering)
GPA: 3.49/4.0

Kurukshetra University, Kurukshetra, India **June 2019**
Bachelor of Technology, Electronics and Communications Engineering
1st Division with Honours, GPA: 4.0/4.0

Professional Experience

University of Pennsylvania, Philadelphia, USA
Robotics Research Engineer, Figueroa GRASP Lab **Jan 2026 – Present**

- Saved \$30,000 in hardware costs and eliminated external tactile skins by engineering a virtual skin manifold for full-body contact detection, achieving 0.2 mm precision at 100 Hz via particle filtering over internal joint torque sensor streams on humanoid hardware.
- Designing a GPU-accelerated voxel-based 3D shape learning pipeline in PyTorch to infer optimal grasping and manipulation strategies for unknown objects in blind state, targeting real-time inference on robot hardware.

Reinforcement Learning ML Researcher, Mod Lab **Aug 2025 – Present**

- Architecting a Proximal Policy Optimization (PPO) reinforcement learning policy π to drive autonomous locomotion adaptation for a quadruped robot generalizing across dynamically mutating morphologies and terrain conditions in simulation (Isaac Lab / MuJoCo).

Research Assistant, Quantum Hardware Lab **Aug 2025 – Dec 2025**

- Engineered an automated quantum control software pipeline for a room-temperature liquid-state NMR system, programmatically sweeping radio-frequency (RF) pulses to map Rabi chevrons on the nuclear spins of a water molecule.
- Accelerated physical qubit calibration time by 80% by extracting optimal π -pulse durations and zero-detuning frequencies from chevron interference patterns, bridging high-level algorithmic intent with low-level hardware execution.

Lead Robotics Engineer, Search and Rescue Autonomy (SMP) **May 2025 – Aug 2025**

- Commissioned and deployed a full-stack autonomous ground navigation suite for Search and Rescue missions, integrating LiDAR, IMU, and wheel odometry sensors with Extended Kalman Filter (EKF) for multi-object tracking and localization, validated in field conditions.
- Engineered a hierarchical planning architecture and tuned a Model Predictive Control (MPC) stack for real-time dynamic obstacle avoidance, achieving 50 Hz closed-loop performance.

Teaching Assistant / Grading Assistant, Engineering Electromagnetics (ESE 2300) **Jan 2025 – May 2025**

- Member of teaching staff for Engineering Electromagnetics; conducted office hours, graded assignments and exams, and delivered recitation sessions covering EM wave theory, Maxwell's equations, magnetism, and antenna fundamentals.

UHG Optum, India **March 2022 – Jul 2024**
Software Engineer, CTO Acceleration Team (Fortune 5 Digital Infrastructure)

- Architected a highly concurrent data ingestion backend in Scala 3 using Akka Actors and Akka Streams, scaling to process 8–11 TB of daily streaming data across $\sim 500\text{M}+$ daily events; reduced cross-cluster P99 processing latency by 30% under heterogeneous cluster load.

- Engineered fault-tolerant Kafka pipelines spanning 20+ topics across 3 consumer groups, implementing at-least-once delivery guarantees, idempotent consumers, and automated partition rebalancing, eliminating data loss under broker failover and reducing consumer lag incidents by 60%.
- Spearheaded migration to a zero-human-touch deployment model, reducing deployment overhead by 90% by architecting automated GitHub Actions CI/CD pipelines and integrating policy-enforcement bots (bulldozer, policy-bot).
- Built a Grafana + CloudWatch observability stack defining 15+ SLO alert rules, P99 latency dashboards, and Kafka consumer lag monitors, enabling on-call teams to detect and resolve production incidents 40% faster.
- Led design and enterprise-wide adoption of a common configuration versioning library (packaged via Maven/Ivy), standardizing disaster recovery and automated AWS S3 backups across multiple engineering squads; implemented content-addressable storage using SHA-256 achieving 35% storage reduction and 3x parallel I/O throughput.
- Deployed and tuned EKS-based microservice workloads on AWS, configuring HPA policies, rolling update strategies, and IAM/VPC security boundaries; provisioned SQS dead-letter queues and MSK clusters to decouple ingestion stages and absorb upstream traffic spikes.
- Integrated Scala-Steward for automatic dependency PRs, eliminating manual library update overhead across all Scala Play back-end micro-services.

HCL Technologies, India

Jul 2019 – March 2022

Software Engineer

- Drove a 48% improvement in operational efficiency for Deutsche Bank’s international cheque clearing by leading integration of a transformer-based human script recognition ML engine into core enterprise systems.
- Architected and deployed highly scalable Java SpringBoot microservices for automated document processing, handling thousands of daily document batches and reducing manual operational effort by 80%.
- Directed end-to-end development of reactive financial web applications using Angular 10 and Spring Boot 4, integrating payment gateways, CRM tools, and AWS cloud services to support complex financial workflows.
- Performed a POC for TWAIN-based web scanning and integrated I/O peripherals (printer/scanner) using Java, Spring Boot, NLP, and Azure for a multi-tenant insurance platform serving Deutsche Bank.
- Built multi-tenant architecture-based software using component inheritance, modularity, and DRY principles for insurance and investment products backed by MySQL and Angular 10.

Projects

Agentic AI Manager (Scala / Akka)

2025

- Improved system resource utilization by 40% and reduced P99 message latency by 150 ms by building a highly scalable Actor system manager on Scala Play, enforcing at-most-once and at-least-once delivery semantics per workload priority tier.
- Designed an intelligent agent selector and load balancer with asynchronous, queue-backed communication protocols to route cross-model distributed workloads across available agents, each running on its own Akka actor.

Heterogeneous Computing Pipeline for Stream Compression (C++ / OpenCL / FPGA) SOC 2025

- Authored a custom OpenCL compute kernel for stream compression deployed on a Xilinx Ultra96 FPGA platform, achieving 135 Mbps per thread throughput; built the C++ host runtime managing host-to-device buffer transfers, kernel dispatch scheduling, and memory coherency.
- Profiled kernel occupancy, memory bandwidth, and host-device transfer overhead; applied tiling and buffer reuse strategies to maximize compute efficiency on the accelerator.
- Maximized horizontal scaling by implementing content-defined chunking (Gear hash table) and ac-

celerating SHA-256 computations using CPU NEON intrinsics.

**Autonomous F1/10th Robo Racers Car (ROS2 / NVIDIA Jetson)
University of Pennsylvania X-Lab 2025**

University of

- Team Lead managing 4 students; deployed a fully autonomous ground vehicle on NVIDIA Jetson, integrating LiDAR, depth camera, and IMU sensors via ROS2 on Ubuntu Linux.
- Implemented sampling-based MPPI Model Predictive Control for steering and throttle and Particle Filter SLAM for localization; achieved 50 Hz hard real-time autonomous control. Placed 2nd in University Competition.
- Built kinematic models, curvature controllers, and sensitive path planners; implemented Kalman Filters, Markovian estimation, and safety-critical multithreaded ROS2 nodes for inter-process communication and real-time scheduling.
- Integrated transformer-based (LoFTR) feature extraction for dense visual correspondence and GPU-accelerated MPPI on CUDA for perception-to-control latency reduction.

Optical Flow and Scene Reconstruction

University of Pennsylvania 2025

- Improved ego-motion tracking accuracy by 7% by implementing Lucas-Kanade optical flow with RANSAC outlier rejection for robust visual localization in challenging lighting and texture conditions.
- Integrated transformer-based (LoFTR) feature extraction for dense visual correspondence enabling robust 3D scene reconstruction from monocular sequences.

RISC-V Microprocessor on Lattice FPGA (Verilog) University of Pennsylvania Jan 2025

- Built a RISC-V ISA compliant microprocessor achieving 17.65 MHz clock speed by implementing a 5-stage pipeline and a custom 8-stage Divider Module; added multicycle execution paths and an L1 cache hierarchy.
- Synthesized and deployed on Lattice FPGA using Xilinx Vivado; validated functional correctness using the Dhrystone benchmark suite with CocoTb simulation framework.

ESE-5090 Quantum Circuits and Systems (IBM Quantum Cloud / Qiskit) University of Pennsylvania 2024–2025

- Implemented and executed Grover's Algorithm, Shor's Algorithm, Quantum Fourier Transform (QFT), Bernstein-Vazirani Algorithm, Simon's Algorithm, and quantum cryptography protocols on IBM Quantum Cloud hardware using Qiskit.
- Analyzed noise, decoherence, and gate fidelity characteristics of real quantum hardware and compared results against ideal circuit simulation.

**Low Light Image Enhancement using Transformer-based Neural Network (RetinexFormer)
University of Pennsylvania Nov 2024**

- Trained a RetinexFormer (transformer) model for low-light image and video enhancement based on Retinex theory, achieving PSNR of 22.7 with live video processing at 5 FPS.
- Applied model acceleration via quantization and pruning and GPU-based vertical acceleration to meet the real-time FPS target; visualized training dynamics using TensorBoard.
- Experimented with Charbonnier Loss, L1-Luminance Loss, Gradient Graph Laplacian Regularizer, ReduceLROnPlateau scheduler, and Lion optimizer, reducing convergence training time by 1.5%.

Deep Neural Network Compression, Quantization & GPU Inference Profiling University of Pennsylvania Nov 2024

- Achieved a 75% reduction in model memory footprint (500 MB to 125 MB) by implementing INT8 quantization and structured pruning on VGG16, lowering the PyTorch computational graph to an ONNX intermediate representation for framework-agnostic deployment.
- Developed a custom sparse Linear layer for iterative pruning and saved the final compressed model in CSR format; experimented with Global, Channel, Structured, and Unstructured pruning variations using PyTorch.

- Profiled inference across CUDA, Apple MPS, and CPU execution providers using ONNX Runtime (ORT), identifying memory-bandwidth bottlenecks per operator and quantifying numerical accuracy degradation under strict hardware constraints.

URL Shortify (Node.js / Express / MongoDB)

Oct 2022

- Designed and built a RESTful Node.js Express back-end microservice with MongoDB for URL shortening; developed a date-based algorithm for generating hexadecimal hashes as redirect identifiers.

Distributed Flight Check-in System (Java / Spring Boot / RabbitMQ)

Jan 2021

- Architected a fault-tolerant distributed system in Java Spring Boot and Oracle SQL, decoupling User, Flight Data, and Check-in services across independently deployable microservices to eliminate single points of failure.
- Guaranteed exactly-once check-in semantics by implementing RabbitMQ messaging queues with idempotent consumers and dead-letter queue routing, achieving 100% reliable asynchronous processing under upstream failure injection.

Social App for Image Sharing (MEAN Stack / AWS S3)

June 2020

- Engineered a full end-to-end MEAN stack application with image blob storage in AWS S3 and a NoSQL MongoDB database for scalable user feed management; Angular client UI with Node/Express back-end deployed on Heroku.

Design and Implementation of Pulse Oximeter for Patient Monitoring (Embedded / Altium)

Oct 2018 – Mar 2019

- Led a 4-member team to design and implement a non-invasive blood oxygen SpO2 monitor using the MAX30100 IR sensor on an Arduino Uno platform; designed a 2-layer PCB schematic in Altium Designer and fabricated the circuit in the micro-electronics lab.
- Developed a companion Android application for user data logging and real-time SpO2 visualization; benchmarked calibration accuracy against an industrial ICU patient monitoring sensor using Digital Signal Processing.
- Researched and applied communication protocols: ADC, PWM, SPI, I2C, Ethernet, and CAN.

Weather Box (Arduino / Firebase / BLE)

June 2017

- Built an IoT weather monitoring box integrating a humidity and temperature sensor (DHT) with an Arduino Uno, Wi-Fi shield, and BLE module; logged live sensor data to Firebase Realtime Database when connected via Wi-Fi.
- Developed a companion Android application for viewing live weather data from Firebase or directly via Bluetooth.

Technical Skills

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- **Languages:** C, C++, CUDA, Python, Scala, Java, JavaScript, TypeScript, Verilog, MATLAB, Haskell, Bash, HTML, SQL, JSON
 - **AI / ML:** PyTorch, TensorFlow, TensorRT, ONNX Runtime, OpenCV, Quantization (INT8/FP16), Structured Pruning, Deep Neural Networks, Transformers, Reinforcement Learning (PPO), Computer Vision
 - **Robotics / Autonomy:** ROS2, SLAM, Particle Filter, EKF, MPC / MPPI, PID Tuning, Path Planning, Isaac Lab, MuJoCo, Rviz, LiDAR, Depth Camera, IMU, Radar, GNSS
 - **Systems / HPC:** CUDA Kernels, OpenCL Kernels, GPU Architecture, Heterogeneous Computing, FPGA (Xilinx Ultra96, Vivado), NVIDIA Jetson, RISC-V, ARM, RTOS, Multithreading, Memory Synchronization
 - **Backend / Distributed:** Spring Boot, Play, Akka, Akka Streams, Kafka (topic mgmt, consumer groups, partition rebalancing), RabbitMQ, Node.js, Express, Microservices, REST, gRPC, Functional Programming
 - **Cloud / DevOps:** AWS (S3, SQS, MSK/Kafka, EKS, IAM, VPC, Lambda), Azure, GCP, Docker,

Kubernetes (HPA, rolling deploys), GitHub Actions CI/CD, Linux, Qiskit (IBM Quantum)

- **Databases:** MongoDB, MySQL, Oracle SQL, PostgreSQL, Firebase
- **Observability:** Prometheus, Grafana, CloudWatch, distributed tracing, SLO/SLA alerting
- **Frontend:** Angular, React, HTML/CSS, TypeScript
- **Hardware / Embedded:** Altium Designer, PCB Design, Arduino, Embedded C, ADC/PWM/SPI/I2C/CAN, Antenna Design (Mentor Graphics), Semiconductor Physics

Certifications and Courses

- **Algorithmic Toolbox**, Coursera Aug 2019
Topics: Data Structures, Greedy Algorithms, Dynamic Programming
- **Angular: The Complete Guide**, Udemy Aug 2019
- **Ethical Hacking Workshop**, Indian Institute of Technology Roorkee Sept 2018
- **Cyber Security Basics: A Hands-on Approach**, Universidad Carlos III de Madrid, edX July 2018
- **Introduction to Linux**, The Linux Foundation, edX July 2018
- **Dual 4-Week Summer Training** in Embedded System Design on Arduino Platform and Android Application Development using Java, Netmax Technologies 2017
- **Summer Training in Embedded Systems & IC Fabrication Techniques**, ACE & App Research July 2016
Topics: IC Fabrication demonstration, embedded circuit design

Conferences, Seminars, and Workshops

- Active participant in **GRASP Lab** (General Robotics, Automation, Sensing & Perception) seminars and colloquiums at University of Pennsylvania; attended invited talks including:
Cal Tech Prof. Pietro Perona on visual perception and recognition systems
Nobel Prize MIT Professor Moungi Bawendi on quantum dots and nanocrystals
- Attended ESE Seminars at University of Pennsylvania covering topics in electrical systems and embedded engineering.
- Attended invited talk on “**Nanocarbon Interconnects**” by Prof. Cary Y. Yang, Santa Clara University, USA, organized by Dept. ECE of Ambala College of Engineering & Applied Research May 2019
- Attended workshop on “**LabVIEW with myDAQ & myRIO**” at National Instruments Centre of Excellence, MMU Oct 2016

Extra-Curriculars and Achievements

- Selected for a 5-day Services Selection Board interview at **18 SSB Allahabad** after clearing the prestigious Indian National UPSC NDA written examination and screening 2015
- **Won 1st prize** in Group Discussion, Picture Perception, and Technical Writing events organized by the IETE Society 2017
- **Runner-up (Technical)** at Inter-collegiate Quiz organized by IETE at CGC Landran Mohali, Chandigarh University 2018
- Participated in **District Level Science Quiz Competition** at Govt. Post Graduate College Ambala 2018–19
- Served as a **member of the Institute of Electronics and Telecommunications Engineers (IETE)** 2017
- Organized a **Blood Donation Camp** in collaboration with the Rotary Club, Ambala, at college 2015–16